Malka Wickramarathna

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Education

University of Moratuwa *2025 – 2029 (Expected)*
B.Sc. (Hons.) Electronic and Telecommunication Engineering

- Current GPA: **4.0/4.0**
- **Key Coursework:** Circuits, Signals and Systems (A+), Introductory Telecommunications (A+), Introductory Electronics (A+), Mathematical Analysis, ODEs and Vectors (A+), Electrical Fundamentals (A+), Programming Fundamentals (A+), Mechanics (A+), Fluid Mechanics (A+), Properties of Materials (A+), Communication Skills (A+)

Royal College, Colombo *2022 - 2024*

- **G.C.E. Advanced Level 2023:** Ranked **5th in District** and **14th in Island** with Distinctions (A's) in Mathematics, Physics, Chemistry, and General English.

D.S. Senanayake College, Colombo *2015 - 2021*

- **G.C.E. Ordinary Level 2020:** Achieved Distinctions in all 9 subjects (9A's).

Technical Projects

Electrocoagulation-based Household Water Purifier Prototype *Jan 2026 - Jul 2026*
Led the Group Project (Engineering Design Project) of 4 members

- Conceptualized a novel, affordable water purification system targeted at affected regions by excess amounts of Hardness, Fluoride, Heavy Metals, and Organic Pollutants in groundwater.
- Led the broad design of the water flow, physical placement of subsystems, and the structural rigidity of the system, considering expert advice and guidance.
- Built a proof-of-concept TRL-4 prototype and validated its functionality using synthetic groundwater in a laboratory setting.

Project Phoenix (CubeSatellite) *Oct 2025 - Jan 2026*
Electronic Club of UoM in collaboration with SEDS Mora

- Developed the Ground Station User Interface (UI) for a high-altitude weather balloon using **Python** and **HTML** for real-time telemetry visualization.
- Collaborated in the validation and testing of the **LoRa-based** communication sub-system.

Smart Hiking Pole *Sept 2025 - Dec 2025*
Group project of 4 members

- Designed an **ESP32-based** smart hiking pole integrating **GPS** and **IMU** sensors for location tracking in extreme environments.

- Developed a path-retracing algorithm utilizing logged GPS coordinates to generate return routes for users.
- Engineered a functional breadboard prototype, addressing technical challenges in signal acquisition and integration.

Skills

- **Programming & Scripting:** Python (Proficient), C/C++, LaTeX, HTML
- **Engineering Software:** MATLAB, Altium Designer
- **Languages:** English (Proficient), Sinhala (Native)

Selected Awards & Honors

G.C.E. Advanced Level Examination Achievements *2023*

- **Presidential Appreciation Award:** Awarded by the President of Sri Lanka to the highest-ranking students in the country for demonstrating exceptional academic aptitude (District Rank 5).
- **Dr. J. Nanayakkara Memorial Prize:** Awarded for achieving the second-highest score in Physics within the 2023 Advanced Level cohort at Royal College, Colombo.
- **Cameron Samarasinghe Memorial Prize:** Awarded for outstanding performance in Combined Mathematics.
- **Royal College OBA (North America):**
  - **2nd Place in Physical Science Stream:** Awarded for outstanding performance in the Physical Science stream.
  - **3rd Place in Overall Performance:** Awarded for consistent academic excellence across all streams.

Professional Affiliations

IEEE Student Member *Since 2026*

Interests

- Classical Piano
- Swimming